Assignment

accredian
credentials that matter

Instructor Effectiveness Modeling (EdTech Context)

Role: Data Science / AI Content Specialist Intern
Submission Format: Google Colab or Jupyter Notebook
Allowed Libraries: Python, pandas, numpy, scikit-learn, matplotlib / seaborn
Not Allowed: LLMs, external datasets, AutoML tools

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1. Problem Context

You are working with an **EdTech platform** that runs the *same course across multiple batches* taught by different instructors.

Each instructor may teach:

- Multiple batches
- The same course across different batches
- Different courses across time

The company wants to better understand **instructor effectiveness** using available data from learner outcomes, engagement, and feedback.

Your task is to **analyze the data, define instructor effectiveness, and build an ML model** to predict instructor effectiveness tiers.

There is **no single correct answer**.
We are more interested in **how you think** than the final model accuracy.

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2. Dataset Description

You are provided with a CSV file where:

- **Each row represents one course batch**
- Multiple rows may belong to the same instructor

2.1 Identifier Columns

Column	Description
batch_id	Unique ID for a course batch
instructor_id	Unique instructor identifier
course_id	Course identifier

2.2 Learner Outcome Metrics

Column	Description
completion_rate	Fraction of learners who completed the course (0–1)
dropout_rate	Fraction of learners who dropped out (0–1)
avg_score_improvement	Average improvement from pre- to post-assessment

avg_quiz_score	Average quiz score for the batch

2.3 Engagement Metrics

Column	Description
avg_watch_time	Normalized average video watch time (0–1)
assignment_submission_rate	Fraction of learners submitting assignments
forum_activity_rate	Fraction of learners active on discussion forums

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2.4 Feedback Metrics

Column	Description
avg_feedback_score	Average learner feedback rating (1–5)
feedback_response_rate	Fraction of learners who submitted feedback

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3. Your Objective

Your overall goal is to **estimate instructor effectiveness** using machine learning.

You will need to:

1. Define what “instructor effectiveness” means
2. Aggregate batch-level data to instructor-level
3. Build an ML model to predict instructor effectiveness tiers
4. Analyze and interpret your results

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4. Tasks to Complete

Step 1: Exploratory Data Analysis (EDA)

- Explore distributions and relationships between variables
- Identify correlations, missing values, or anomalies
- Highlight any early observations

📌 *Use plots and short explanations.*

Step 2: Define Instructor Effectiveness

You must define an **Instructor Effectiveness Score** using the available data.

Then convert this score into **effectiveness tiers**, for example:

- Low
- Medium
- High

⚠️ There is no predefined formula.

You must **justify your definition** and assumptions.

Step 3: Aggregate Batch Data to Instructor Level

Since instructors teach multiple batches:

- Aggregate batch-level features into instructor-level features
- Decide how to handle instructors with:
    - Few batches
    - Many batches

Explain:

- Which aggregation functions you used
- Why they make sense

Step 4: Build a Machine Learning Model

- Train a model to predict instructor effectiveness tier
- You may choose any **classical ML algorithm**
- Feature selection and scaling are optional but encouraged

📌 *Model choice matters less than your reasoning.*

Step 5: Evaluate the Model

- Choose appropriate evaluation metrics

- Discuss trade-offs (e.g., precision vs recall)
- Comment on class imbalance if present

Step 6: Interpret the Results

- Identify key features influencing instructor effectiveness
- Explain results in simple, non-technical terms
- Suggest how insights could be used in an EdTech product

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5. Mandatory Analysis Questions

(Answer in markdown cells in the notebook)

1. Which features most influenced instructor effectiveness, and why?
2. Which variables could be misleading or confounded?
3. How could this model fail in real-world usage?
4. What additional data would you want to improve this analysis?
5. Should this model be used for instructor performance evaluation? Why or why not?

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6. Constraints & Guidelines

- ❌ Do not use LLMs or external APIs
- ❌ Do not use external datasets
- ✅ Focus on clarity and reasoning
- ✅ Keep the notebook reproducible
- ✅ Comment your code and decisions

Accuracy is **not the primary evaluation metric**.

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7. What We Are Evaluating

We will assess your submission on:

- Understanding of the problem
- Quality of feature engineering & aggregation
- ML fundamentals and evaluation logic
- Interpretation and communication
- Awareness of limitations and ethical considerations

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8. Submission Instructions

- Submit a **single notebook file**
- Ensure all cells run without errors
- Include explanations in markdown cells
- Clearly label sections

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9. Final Note to Candidates

This task reflects real problems faced by EdTech companies.

Strong submissions:

- Clearly explain assumptions
- Discuss trade-offs
- Acknowledge limitations
- Demonstrate structured thinking

There is **no penalty for a simple model** if the reasoning is strong.

Best of luck — we're excited to see how you think.